



KG-2 BOOK B



Decomposition of 10 & Big/Small Friends

Student Information

Name:

.....

Level:

.....

Date:

.....

8

9

5



Combination of Big & Small Friends

Teen – single rewrites as – 10 + (10 – single)



★

$$13 - 8 : 13 - 10 + 2 = ()$$



✱

$$\begin{array}{r} 13 \\ - 8 \\ \hline \end{array}$$

★

☰

☰

$$14 - 9 : 14 - 10 + 1 = ()$$



$$\begin{array}{r} 14 \\ - 9 \\ \hline \end{array}$$



Decomposition Pyramid

 Solve every subtraction problem in the staircase



10-1									
10-2	11-2								
10-3	11-3	12-3							
10-4	11-4	12-4	13-4						
10-5	11-5	12-5	13-5	14-5					
10-6	11-6	12-6	13-6	14-6	15-6				
10-7	11-7	12-7	13-7	14-7	15-7	16-7			
10-8	11-8	12-8	13-8	14-8	15-8	16-8	17-8		
10-9	11-9	12-9	13-9	14-9	15-9	16-9	17-9	18-9	



Abacus Calculation: Decomposition of 10 (–9)



–9 = –10 + 1 Fingering Ex.: 10–9, 11–9, 12–9

S.No.	1	2	3	4	5	6	7	8
1	10	11	12	13	10	11	12	13
2	-9	-9	-9	-9	-9	-9	-9	-9
3	2	3	4	1	5	2	3	4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	14	15	16	17	18	11	12	13
2	-9	-9	-9	-9	-9	-9	-9	-9
3	3	2	1	3	2	4	5	6
Ans.								

S.No.	1	2	3	4
1	14	15	16	17
2	-9	-9	-9	-9
3	2	3	4	1
Ans.				



11 – 9 =



12 – 9 =



13 – 9 =

13 – 9

= 13-10+1

= 4



Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	12	13	14	15	16	17	10	11
2	-9	-9	-9	-9	-9	-9	-9	-9
3	3	5	2	4	3	2	4	3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	18	12	13	14	15	16	10	11
2	-9	-9	-9	-9	-9	-9	-9	-9
3	1	5	2	3	4	2	3	5
Ans.								

S.No.	1	2	3	4	☆
1	12	13	14	15	
2	-9	-9	-9	-9	
3	4	3	2	1	
Ans.					



$11 - 8 =$



$13 - 7 =$



$15 - 6 =$



$12 - 9 =$

Abacus Calculation: Decomposition of 10 (–8)



–8 = –10 + 2 Fingering Ex.: 11–8, 13–8, 15–8, 16–8

S.No.	1	2	3	4	5	6	7	8
1	11	12	13	14	15	16	17	18
2	-8	-8	-8	-8	-8	-8	-8	-8
3	3	4	2	5	3	2	4	1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	10	12	14	16	11	13	15	17
2	-8	-8	-8	-8	-8	-8	-8	-8
3	5	3	2	4	6	1	5	3
Ans.								

S.No.	1	2	3	4
1	10	12	14	16
2	-8	-8	-8	-8
3	4	2	3	1
Ans.				

11 – 8 =

13 – 8 =

14 – 8 =

11 – 8
= 11-10+2
= 3

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	12	10	11	14	16	15	13	12
2	-9	-8	-9	-8	-9	-8	-9	-8
3	3	4	2	3	4	2	3	5
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	17	16	15	14	13	12	11	10
2	-9	-8	-9	-8	-9	-8	-9	-8
3	2	3	4	1	5	4	3	2
Ans.								

S.No.	1	2	3	4
1	18	16	14	12
2	-9	-8	-9	-8
3	3	1	4	2
Ans.				

$15 - 7 =$

$14 - 6 =$

$11 - 5 =$

$12 - 4 =$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	11	12	13	14	15	16	17	18
2	-8	-9	-8	-9	-8	-9	-8	-9
3	3	2	4	1	3	4	2	3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	10	11	12	13	14	15	16	17
2	-9	-8	-9	-8	-9	-8	-9	-8
3	4	5	3	2	4	1	3	5
Ans.								

S.No.	1	2	3	4
1	10	12	14	16
2	-8	-9	-8	-9
3	3	2	4	1
Ans.				

13 – 4 =

16 – 9 =

15 – 8 =

14 – 7 =

Abacus Calculation: Decomposition of 10 (-7)



-7 = -10 + 3 Fingering Ex.: 11-7, 15-7, 16-7

S.No.	1	2	3	4	5	6	7	8
1	11	12	13	14	15	16	17	18
2	-7	-7	-7	-7	-7	-7	-7	-7
3	4	3	2	5	3	2	4	1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	10	12	14	16	11	13	15	17
2	-7	-7	-7	-7	-7	-7	-7	-7
3	5	3	2	4	6	1	5	3
Ans.								

S.No.	1	2	3	4
1	10	12	14	16
2	-7	-7	-7	-7
3	4	2	3	1
Ans.				

11 - 7 =

13 - 7 =

15 - 7 =

11 - 7
= 11-10+3
= 4

Visual Subtraction

Subtract 6 from each — what is left?



$$11 - 6 = \boxed{}$$



$$13 - 6 = \boxed{}$$



$$16 - 6 = \boxed{}$$



$$15 - 6 = \boxed{}$$

Visual Subtraction

Different starting numbers, same trick



$$10 - 6 : 10 - 10 + 4 = ()$$



$$\begin{array}{r} 10 \\ - 6 \\ \hline \end{array}$$

$$15 - 6 : 15 - 10 + 4 = ()$$



$$\begin{array}{r} 15 \\ - 6 \\ \hline \end{array}$$



Abacus Calculation: Decomposition of 10 (–6)



–6 = –10 + 4 Fingering Ex.: 10–6, 15–6, 16–6

S.No.	1	2	3	4	5	6	7	8
1	10	11	12	13	14	15	16	17
2	-6	-6	-6	-6	-6	-6	-6	-6
3	2	4	3	5	1	3	2	4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	18	12	13	14	15	16	10	11
2	-6	-6	-6	-6	-6	-6	-6	-6
3	5	3	2	5	1	3	4	2
Ans.								

S.No.	1	2	3	4
1	12	13	14	15
2	-6	-6	-6	-6
3	5	4	2	3
Ans.				

10 – 6 =

12 – 6 =

14 – 6 =

15 – 6
= 15-10+4
= 9

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	15	16	17	18	14	13	12	11
2	-6	-7	-8	-9	-6	-7	-8	-9
3	3	4	2	1	5	3	4	2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	10	11	12	13	14	15	16	17
2	-6	-7	-8	-9	-6	-7	-8	-9
3	4	3	2	1	5	3	4	2
Ans.								

S.No.	1	2	3	4
1	18	16	14	12
2	-6	-7	-8	-9
3	5	3	4	2
Ans.				

10 – 5 =

12 – 5 =

11 – 4 =

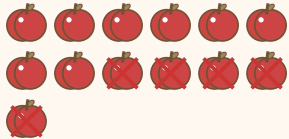
13 – 3 =

Visual Subtraction Cards

Three more subtraction tricks



$$13 - 5 : 13 - 10 + 5$$



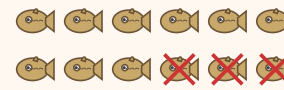
$$\begin{array}{r} 13 \\ - 5 \\ \hline \end{array}$$

$$11 - 4 : 11 - 10 + 6$$



$$\begin{array}{r} 11 \\ - 4 \\ \hline \end{array}$$

$$12 - 3 : 12 - 10 + 7$$



$$\begin{array}{r} 12 \\ - 3 \\ \hline \end{array}$$



Abacus Calculation: Decomposition of 10 (-5)



-5 = -10 + 5 Fingering Ex.: 10-5, 11-5, 12-5

S.No.	1	2	3	4	5	6	7	8
1	10	11	12	13	14	15	16	17
2	-5	-5	-5	-5	-5	-5	-5	-5
3	3	2	4	1	5	3	2	4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	18	10	11	12	13	14	15	16
2	-5	-5	-5	-5	-5	-5	-5	-5
3	3	5	4	3	2	1	5	3
Ans.								

S.No.	1	2	3	4
1	10	12	14	16
2	-5	-5	-5	-5
3	4	2	3	1
Ans.				

$10 - 5 =$

$11 - 5 =$

$12 - 5 =$

$13 - 5$
 $= 13 - 10 + 5$
 $= 8$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8	9	10
1	11	31	20	43	14	35	26	24	41	12
2	-5	-9	-4	-8	-6	-7	-8	-5	-9	-4
3	2	4	3	5	2	1	3	4	2	3
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	13	20	35	14	15	26	18	27	16	19
2	-7	-4	-9	-5	-6	-7	-9	-8	-7	-9
3	5	3	2	1	4	3	5	2	1	4
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	21	24	28	15	16	18	24	14	12	28
2	-6	-7	-9	-8	-7	-9	-5	-7	-9	-7
3	3	2	1	5	2	3	1	4	3	2
Ans.										



Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	12	13	14	15	16	11	12	13
2	-5	-6	-7	-8	-9	-4	-5	-6
3	4	3	2	1	3	5	2	4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	14	15	16	11	12	13	14	15
2	-7	-8	-9	-4	-5	-6	-7	-8
3	3	2	4	3	5	1	4	3
Ans.								

*

S.No.	1	2	3	4
1	16	11	12	13
2	-9	-4	-5	-6
3	2	4	3	2
Ans.				

15 – 3 =

14 – 4 =

13 – 5 =

12 – 6 =

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	12	14	16	11	13	15	17	10
2	-6	-8	-9	-5	-7	-9	-8	-4
3	3	2	4	3	2	4	3	5
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	18	16	14	12	10	11	13	15
2	-9	-8	-7	-6	-5	-4	-7	-9
3	2	4	3	5	2	4	3	2
Ans.								

S.No.	1	2	3	4
1	17	15	13	11
2	-9	-7	-5	-4
3	3	2	4	1
Ans.				



$16 - 7 =$



$15 - 8 =$



$14 - 9 =$



$13 - 4 =$

Abacus Calculation: Decomposition of 10 (-4)



-4 = -10 + 6 Fingering Ex.: 10-4, 11-4, 12-4

S.No.	1	2	3	4	5	6	7	8
1	10	11	12	13	14	15	16	17
2	-4	-4	-4	-4	-4	-4	-4	-4
3	3	5	2	4	3	1	5	2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	18	10	11	12	13	14	15	16
2	-4	-4	-4	-4	-4	-4	-4	-4
3	4	5	3	2	1	5	3	2
Ans.								

S.No.	1	2	3	4
1	10	12	14	16
2	-4	-4	-4	-4
3	4	3	2	1
Ans.				

$10 - 4 =$

$11 - 4 =$

$12 - 4 =$

$13 - 4$
 $= 13 - 10 + 6$
 $= 9$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	24	15	28	11	19	22	17	14
2	-9	4	-7	5	-6	-8	5	3
3	3	-6	5	-9	3	4	-3	-5
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	26	18	15	20	13	21	16	22
2	-9	-8	4	-7	6	-9	3	-7
3	4	3	-2	5	-4	2	-5	4
Ans.								

S.No.	1	2	3	4
1	28	16	24	18
2	-9	5	-8	-6
3	3	-7	2	3
Ans.				

14 – 5 =

15 – 6 =

13 – 4 =

12 – 3 =

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	26	14	18	22	16	28	15	19
2	-9	5	-7	5	4	-9	3	-6
3	3	-6	4	-3	-7	5	-4	2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	24	12	26	15	18	21	23	14
2	-8	4	-9	3	4	-5	-9	5
3	3	-5	2	-6	-7	3	4	-2
Ans.								

S.No.	1	2	3	4
1	20	26	12	28
2	-8	-7	5	-9
3	5	3	-4	1
Ans.				



Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	26	18	24	15	28	17	22	13
2	-9	5	-7	4	-9	5	-8	4
3	3	-6	4	-3	2	-4	3	-6
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	20	14	26	16	24	18	15	27
2	-7	5	-9	3	-8	-6	4	-9
3	3	-2	4	-5	2	3	-7	5
Ans.								

S.No.	1	2	3	4
1	19	23	14	26
2	-8	5	-3	-9
3	3	-7	5	3
Ans.				



Match the Mascot to its House

Each mascot's problem solves to a house number



$$14 - 6$$



$$13 - 5$$



$$12 - 4$$



$$11 - 3$$



$$15 - 7$$



$$16 - 8$$



$$17 - 9$$



$$10 - 2$$



Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	11	24	31	16	19	27	14	28
2	-5	-9	-7	4	5	-9	3	-8
3	3	4	5	-6	-3	4	-5	2
Ans.								



S.No.	1	2	3	4	5	6	7	8
1	12	24	35	15	16	26	18	27
2	-5	-9	-8	4	3	-7	-9	-8
3	4	3	2	-6	-5	3	5	2
Ans.								



S.No.	1	2	3	4
1	21	28	24	15
2	-6	-7	-9	-8
3	3	4	1	5
Ans.				



Count and Subtract

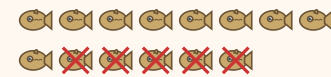
Cross out the smaller group and write the difference



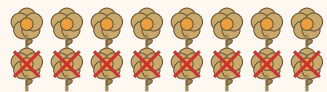
$$12 - 4 = \boxed{}$$



$$15 - 7 = \boxed{}$$



$$14 - 5 = \boxed{}$$



$$16 - 8 = \boxed{}$$



$$13 - 6 = \boxed{}$$



$$11 - 2 = \boxed{}$$



Abacus Calculation: Combination of 10 (–9 two-step)



–9 = –10 + 5 – 4 Fingering Ex.: 14–9

S.No.	1	2	3	4	5	6	7	8
1	14	14	14	14	14	14	14	14
2	-9	-9	-9	-9	-9	-9	-9	-9
3	3	5	2	4	1	3	5	2
* Ans.								

S.No.	1	2	3	4	5	6	7	8
1	14	14	14	14	14	14	14	14
2	-9	-9	-9	-9	-9	-9	-9	-9
3	4	3	2	5	1	4	3	2
Ans.								

S.No.	1	2	3	4
1	14	14	14	14
2	-9	-9	-9	-9
3	5	4	2	3
Ans.				

14 – 9 =

15 – 9 =

16 – 9 =

14 – 9
= 14-10+5-4
= 5

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	14	15	16	14	15	16	14	15
2	-9	-9	-9	-9	-9	-9	-9	-9
3	3	2	4	1	3	5	2	4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	13	14	12	13	14	15	16	14
2	-8	-8	-7	-7	-8	-9	-9	-9
3	2	3	4	3	1	5	3	4
Ans.								

S.No.	1	2	3	4
1	15	14	16	13
2	-9	-9	-9	-8
3	1	3	2	4
Ans.				

$13 - 8 =$

$12 - 7 =$

$14 - 8 =$

$15 - 8 =$

Abacus Calculation: Combination of 10 (–8 two-step)



–8 = –10 + 5 – 3 Fingering Ex.: 13–8, 14–8

S.No.	1	2	3	4	5	6	7	8
1	13	14	13	14	13	14	13	14
2	-8	-8	-8	-8	-8	-8	-8	-8
3	4	3	5	2	1	4	3	5
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	14	13	14	13	14	13	14	13
2	-8	-8	-8	-8	-8	-8	-8	-8
3	2	4	1	3	5	2	4	3
Ans.								

S.No.	1	2	3	4
1	13	14	13	14
2	-8	-8	-8	-8
3	2	5	4	1
Ans.				

13 – 8 =

14 – 8 =

15 – 8 =

13 – 8
= 13-10+5-3
= 5

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	13	14	15	12	13	14	15	16
2	-7	-8	-9	-7	-8	-9	-7	-8
3	3	2	4	3	5	2	4	1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	12	13	14	15	16	13	14	15
2	-7	-8	-9	-7	-8	-9	-7	-8
3	5	3	2	4	3	5	2	3
Ans.								

S.No.	1	2	3	4
1	16	12	14	15
2	-9	-7	-8	-9
3	4	1	5	3
Ans.				

$13 - 7 =$

$14 - 7 =$

$15 - 7 =$

$16 - 7 =$

Abacus Calculation: Combination of 10 (-7 two-step)



-7 = -10 + 5 - 2 Fingering Ex.: 12-7, 13-7

S.No.	1	2	3	4	5	6	7	8
1	12	13	12	13	12	13	12	13
2	-7	-7	-7	-7	-7	-7	-7	-7
3	3	5	4	2	1	4	5	3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	13	12	13	12	13	12	13	12
2	-7	-7	-7	-7	-7	-7	-7	-7
3	2	4	1	3	5	2	4	3
Ans.								

S.No.	1	2	3	4
1	12	13	12	13
2	-7	-7	-7	-7
3	2	5	4	1
Ans.				

12 - 7 =

13 - 7 =

14 - 7 =

12 - 7
= 12-10+5-2
= 5

Abacus Calculation: Combination of 10 (-7 cont.)



-7 = -10 + 5 - 2 Fingering Ex.: 12-7, 13-7

S.No.	1	2	3	4	5	6	7	8
1	12	13	12	13	12	13	12	13
2	-7	-7	-7	-7	-7	-7	-7	-7
3	5	3	2	4	3	5	1	2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	13	12	13	12	13	12	13	12
2	-7	-7	-7	-7	-7	-7	-7	-7
3	4	3	1	4	2	5	3	4
Ans.								

S.No.	1	2	3	4
1	12	13	12	13
2	-7	-7	-7	-7
3	1	4	5	3
Ans.				

12 - 7 =

13 - 7 =

15 - 7 =

16 - 7 =

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	12	11	12	13	14	12	13	14
2	-6	-6	-7	-8	-9	-7	-8	-9
3	4	3	2	5	1	3	4	2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	11	12	13	14	11	12	13	14
2	-6	-7	-8	-9	-6	-7	-8	-9
3	5	4	3	2	1	5	4	3
Ans.								

S.No.	1	2	3	4
1	12	13	14	15
2	-6	-7	-8	-9
3	2	4	3	2
Ans.				


 $11 - 6 =$


 $12 - 6 =$


 $13 - 6 =$


 $15 - 6 =$

Abacus Calculation: Combination of 10 (-6 two-step)



-6 = -10 + 5 - 1 Fingering Ex.: 11-6, 12-6

S.No.	1	2	3	4	5	6	7	8
1	11	12	11	12	11	12	11	12
2	-6	-6	-6	-6	-6	-6	-6	-6
3	3	5	4	2	1	4	5	3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	12	11	12	11	12	11	12	11
2	-6	-6	-6	-6	-6	-6	-6	-6
3	2	4	1	3	5	2	4	3
Ans.								

S.No.	1	2	3	4
1	11	12	11	12
2	-6	-6	-6	-6
3	2	5	4	1
Ans.				



11 - 6 =



12 - 6 =



13 - 6 =

11 - 6
= 11-10+5-1
= 5



Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	12	13	14	15	11	12	13	14
2	-6	-7	-8	-9	-5	-6	-7	-8
3	4	5	3	2	4	3	2	5
Ans.								



S.No.	1	2	3	4	5	6	7	8
1	15	11	12	13	14	15	11	12
2	-9	-5	-6	-7	-8	-9	-5	-6
3	1	3	4	2	5	3	2	4
Ans.								

S.No.	1	2	3	4
1	13	14	15	12
2	-7	-8	-9	-6
3	1	3	5	4
Ans.				



13 – 5 =

14 – 6 =

15 – 7 =

12 – 4 =

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	13	12	11	10	15	14	13	12
2	-9	-8	-7	-6	-9	-8	-7	-6
3	3	4	2	5	3	2	4	1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	11	10	15	14	13	12	11	10
2	-9	-8	-7	-6	-9	-8	-7	-6
3	5	3	2	4	3	5	4	2
Ans.								

S.No.	1	2	3	4
1	15	14	13	12
2	-9	-8	-7	-6
3	1	3	4	5
Ans.				

11 – 7 =

12 – 8 =

13 – 9 =

14 – 5 =

Mental Calculation



S.No.	1	2	3	4	5	6	7	8	9	10
A	2	5	7	4	6	3	8	2	5	7
2	3	-3	-3	2	-2	5	-4	3	-2	-5
3	-1	4	2	-3	3	-2	2	4	1	3
Ans.										
Ans.										



Two-Digit Addition: Carrying Forward

Now let's apply the trick to bigger numbers

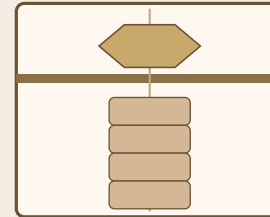


Have you learned how to
carry-forward a digit?
Now do it for double digits!

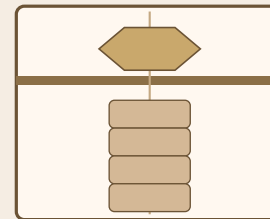
100

1	99
2	98
3	97
4	96
5	95
6	94
7	93
8	92
9	91

$$29 + 61 = 90$$



$$47 + 53 = 100$$



Mental Calculation



S.No.	1	2	3	4	5	6	7	8	9	10
A	3	7	2	8	5	2	8	4	7	5
2	5	-5	4	-6	-2	3	-6	2	-4	3
3	-2	2	3	2	4	5	1	-3	1	-2
Ans.										
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
B	2	7	3	6	2	8	5	4	6	5
2	3	-5	4	3	7	-3	2	-3	2	3
3	2	1	-2	-5	-5	2	1	5	-4	-2
Ans.										
Ans.										

Listening Exercise

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

Final Assessment

Sections B, C, D — show what you have learned



Time: _____

Section B — Abacus Calculation

S.No.	1	2	3	4	5	6	7	8
1	18	21	30	28	17	25	16	12
2	-5	4	-8	-4	6	4	5	9
3	-4	3	-9	6	-7	2	8	-6
Ans.								

Section C — Mental Calculation

S.No.	1	2	3	4	5	6	7	8	9	10
1	9	7	4	6	5	8	2	9	3	6
2	-3	-3	5	2	-2	2	7	-7	9	-5
3	2	5	-2	-1	8	-7	-3	5	-2	9
Ans.										

Section D — Fill in the Combination & Decomposition Number

